

Winning the Space Race with Data Science

Predicting Falcon 9 First-Stage Landing Success

A technical report covering data collection, wrangling, SQL, EDA, geospatial analytics, interactive dashboards, and classification models.

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Software Engineer · Data Science Capstone · 2026

HEADLINE RESULTS

90	Falcon 9 launches analyzed
66.7%	successful landings overall
83.3%	test accuracy (all 4 models)
\$103M	cost gap vs expendable launch

GitHub Repository

REQUIRED: public repository containing all notebooks and Python files

<https://github.com/lelandwrightofficial-alt/testrepo>

Public repository with completed notebooks and Python files. This PDF is the graded submission artifact.

Notebooks in the repository

1. jupyter-labs-spacex-data-collection-api.ipynb
2. jupyter-labs-webscraping.ipynb
3. labs-jupyter-spacex-Data wrangling.ipynb
4. jupyter-labs-eda-dataviz.ipynb
5. jupyter-labs-eda-sql-coursera_sqlite.ipynb
6. lab_jupyter_launch_site_location.ipynb
7. spacex_dash_app.py
8. SpaceX_Machine_Learning_Prediction.ipynb

Why this repo matters

- Reproduces every lab from API collection through classification.
- Includes the Plotly Dash app (Python file) and Folium map notebook.
- Stores wrangled CSVs so reviewers can rerun EDA and models.
- This presentation is the narrative layer over that code.

Presentation Outline

01 Context

- Executive summary
- Business problem
- Why landing prediction matters

02 Methodology

- API + Wikipedia collection
- Wrangling and class labels
- EDA, SQL, Folium, Dash, ML

03 Results

- Visualization findings
- SQL answers
- Maps, dashboard, models

04 Decision

- Best classifier
- Operational insights
- Conclusions and next steps

Executive Summary

THE QUESTION

Can we predict whether a Falcon 9 first stage will land successfully, so a competing launch provider can price a bid against SpaceX's \$62M reusable launch versus ~\$165M expendable launches?

66.7%

Historical landing success across 90 Falcon 9 flights after Falcon 1 removal.

KSC LC-39A

Highest site success rate; CCAFS SLC-40 flew the most missions (55).

Decision Tree

Best validation accuracy (~88.9% CV). All models hit 83.3% on 18 test rows.

2015 turning point

First ground-pad success on 22 Dec 2015; yearly success then climbed toward 80%+.

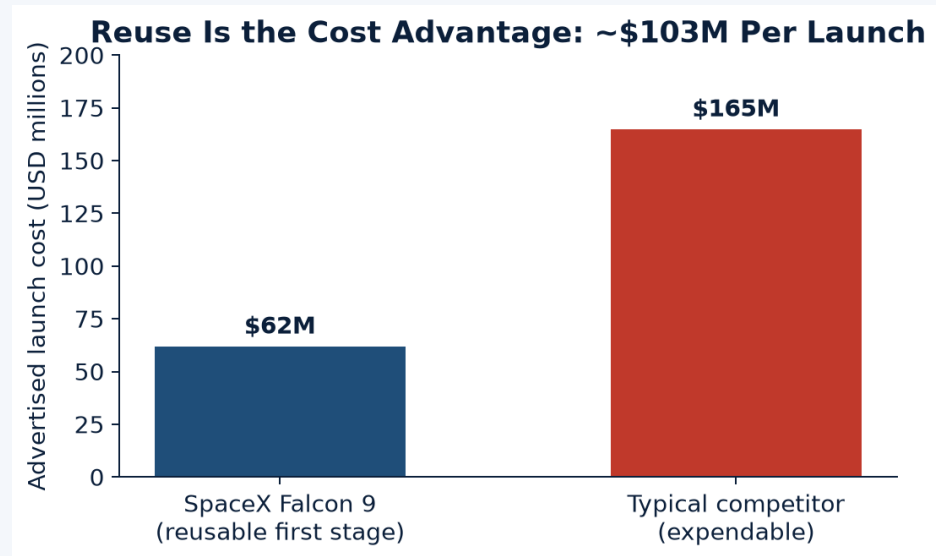
Introduction

The commercial launch market

- SpaceX advertises Falcon 9 launches at about \$62 million.
- Other providers often exceed \$165 million per launch.
- Most of that gap comes from recovering and reusing the first stage.
- If the booster is expended, the economic advantage disappears.

Our role

- We act as data scientists for a startup bidding against SpaceX.
- The product is a landing-success classifier, not a launch vehicle.
- A predicted landing implies a reusable-cost bid; a predicted failure implies expendable-cost planning.
- Audience for this deck: peer data scientists. An executive cut can be the summary + conclusion only.



Methodology Overview

1. Collect

SpaceX REST API static JSON + Wikipedia Falcon 9 table via BeautifulSoup.

2. Wrangle

Filter Falcon 9, impute payload mean, encode Class from Outcome.

3. Explore

Seaborn/Matplotlib EDA, SQLite queries, Folium maps, Plotly Dash.

4. Model

One-hot features, StandardScaler, train/test 80/20, GridSearchCV on 4 classifiers.

5. Decide

Compare CV and test accuracy, inspect confusion matrices, report operational insight.

Data Collection and Data Wrangling

Collection

- API: GET the IBM static launch JSON, decode with `.json()`, flatten with `pd.json_normalize`.
- Enrichment: booster, launch site, payload, and core fields (Outcome, Flights, GridFins, Legs, Block, Serial).
- Web scrape: BeautifulSoup on the Falcon 9 Wikipedia launch table; extract headers and row records.
- SQL source: SpaceX.csv loaded into SQLite table SPACEXTBL.

Wrangling

- Keep one core and one payload per launch; drop Falcon Heavy-style multi-core rows.
- Restrict dates through 13 Nov 2020; remove Falcon 1 → 90 Falcon 9 rows.
- PayloadMass: replace `np.nan` with column mean (~6104.96 kg).
- LandingPad has 26 missing values (no assigned pad / no attempt).
- Class = 0 if Outcome is in `bad_outcomes`, else 1. Success rate = 0.6667.

DATA CONTRACT USED FOR MODELING

Features: FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial

One-hot encode Orbit, LaunchSite, LandingPad, Serial → 80 numeric columns, then `astype('float64')`.

Target Y = `data['Class'].to_numpy()` | Split: `test_size=0.2, random_state=2` → 72 train / 18 test records.

EDA and Interactive Visual Analytics

Statistical EDA

- value_counts on LaunchSite, Orbit, Outcome.
- catplot: FlightNumber vs LaunchSite, hue=Class.
- catplot: PayloadMass vs LaunchSite / Orbit.
- Bar chart of mean(Class) by Orbit.
- Line chart of yearly success rate.
- These plots test whether reuse, site, orbit, and time structure the label.

SQL EDA

- DISTINCT Launch_Site (4 sites).
- LIKE 'CCA%' LIMIT 5.
- SUM payload for NASA (CRS).
- AVG payload for F9 v1.1%.
- MIN Date of Success (ground pad).
- Boosters with drone-ship success and 4000–6000 kg.
- COUNT of Mission_Outcome; MAX payload subquery; 2015 failures; ranked outcomes 2010–2017.

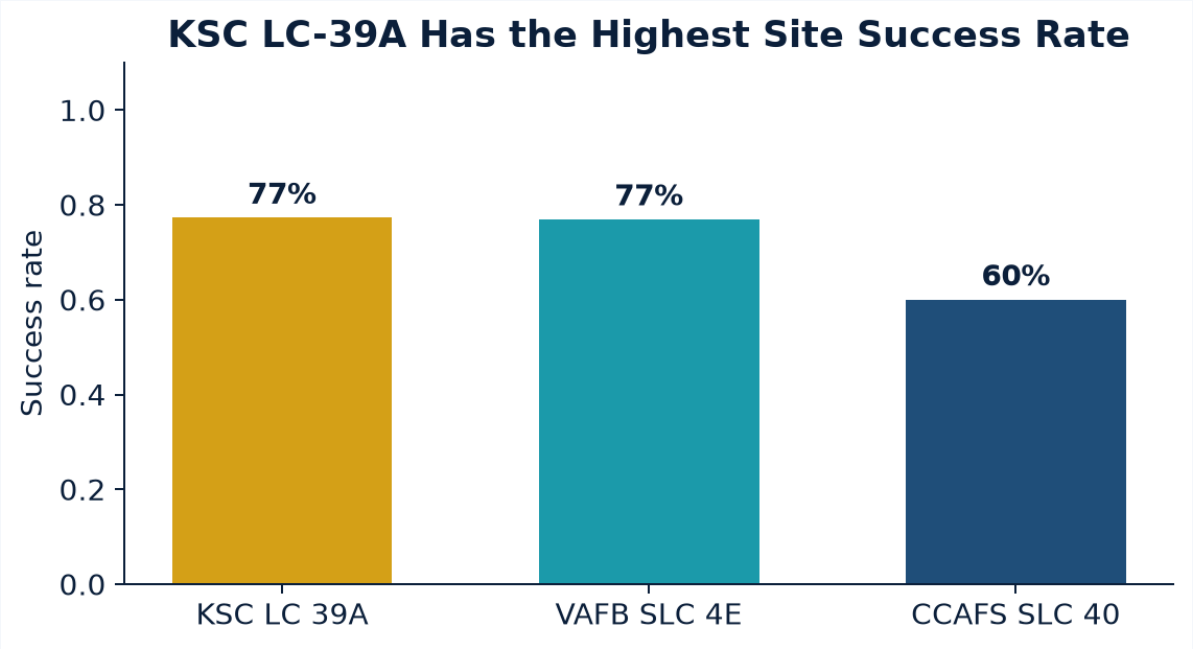
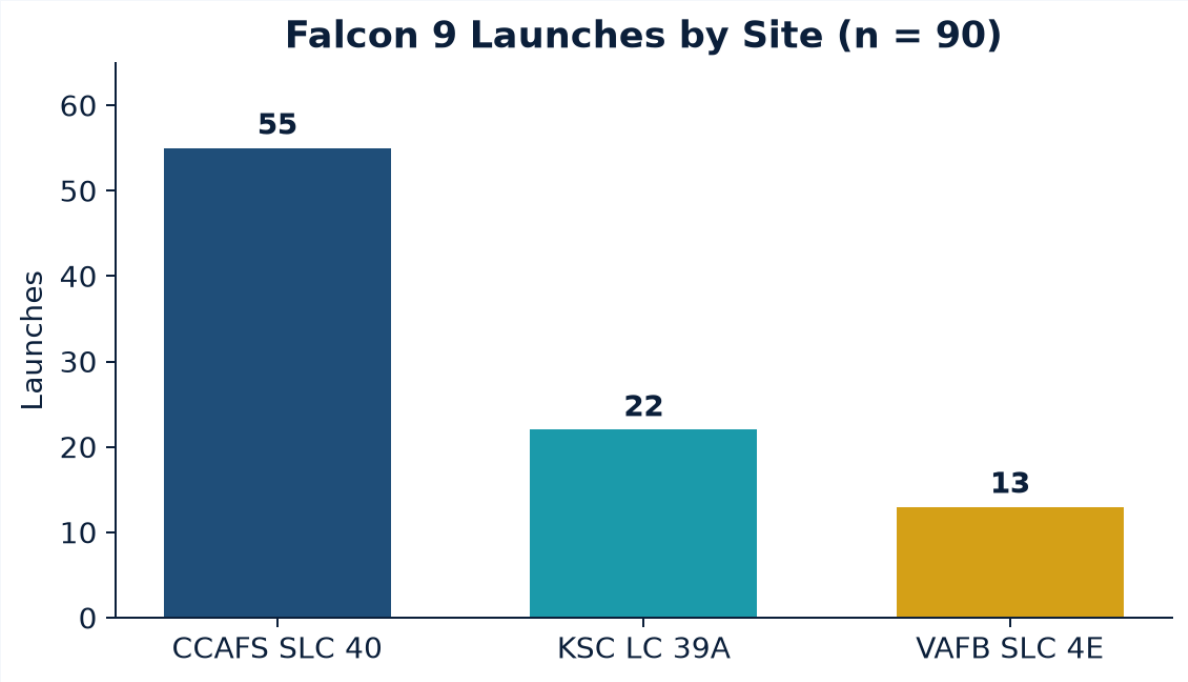
Folium + Dash

- Folium Map + Circle + DivIcon Marker per site.
- MarkerCluster for overlapping launch records; green = Class 1, red = Class 0.
- MousePosition + distance lines to coast, rail, highway, city.
- Dash: site dropdown (options=sites), pie of success, payload scatter, range slider.
- Callbacks bind unique component ids; multiple Input values are allowed.

Predictive Analysis (Classification)

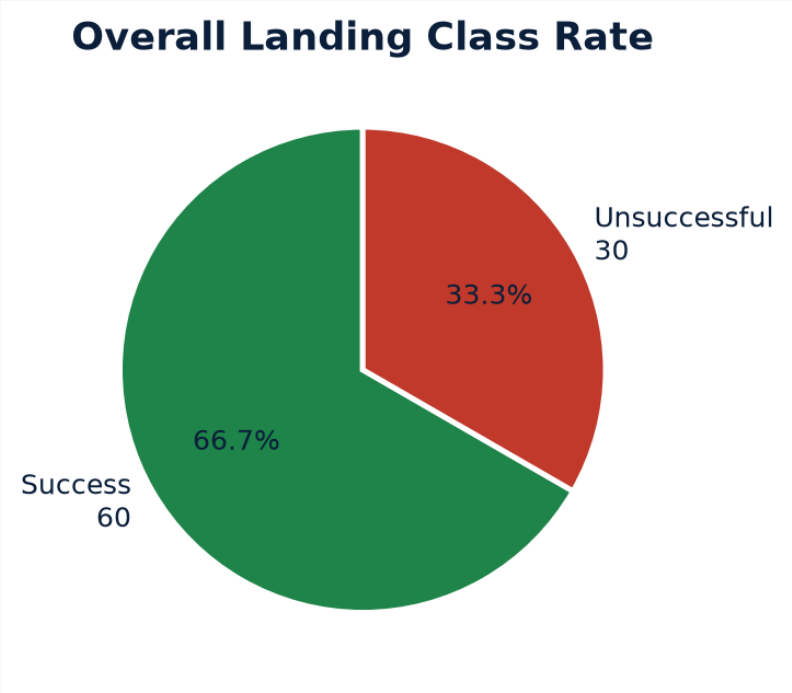
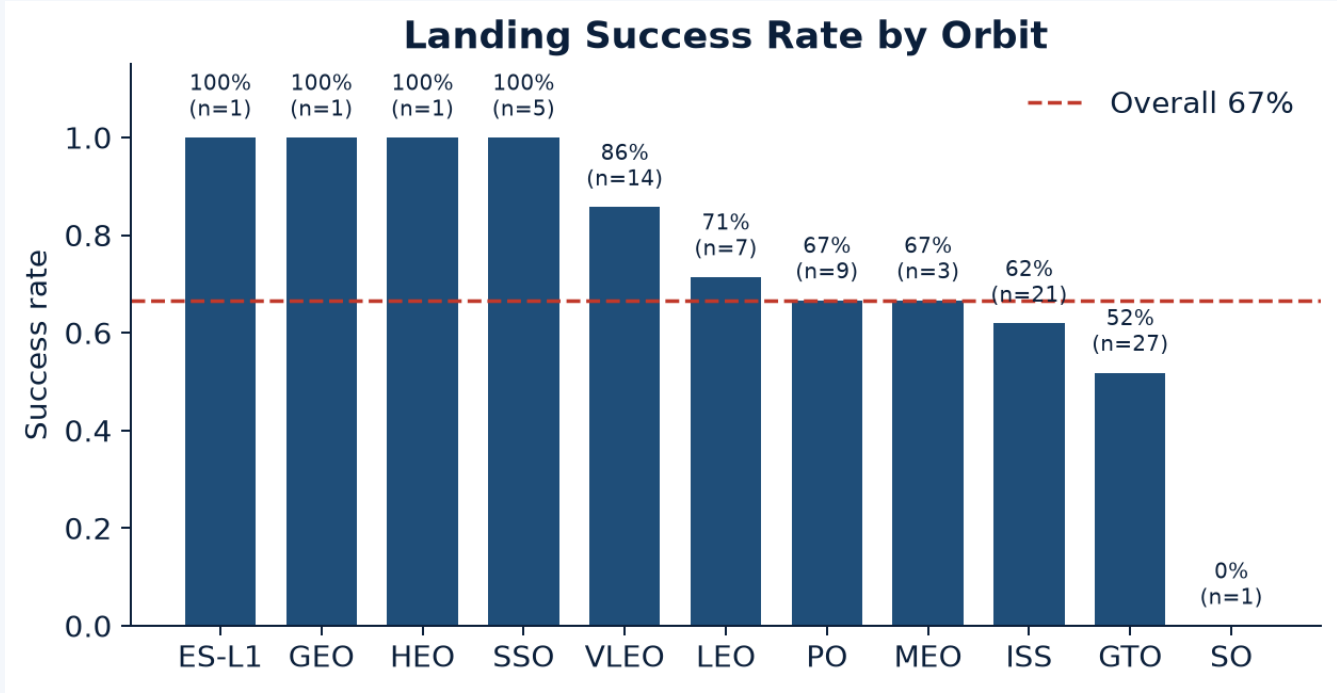
Step	Implementation
Target	Y = data['Class'].to_numpy() (1 = successful landing)
Split	train_test_split(X, Y, test_size=0.2, random_state=2) → 18 test records
Scale	StandardScaler on numeric one-hot matrix (80 columns)
LogReg	C ∈ {0.01, 0.1, 1}, penalty=l2, solver=lbfgs, GridSearchCV cv=10
SVM	kernel ∈ {linear, rbf, poly, sigmoid}, C and gamma logspace(-3,3,5), cv=10
Tree	gini/entropy, best/random splitter, depth 2–18, max_features auto/sqrt, cv=10
KNN	n_neighbors, algorithm, p in {1,2}, GridSearchCV cv=10
Select	Rank by best_score_ (validation) and score(X_test, Y_test); plot confusion matrix

Launch Sites and Landing Class



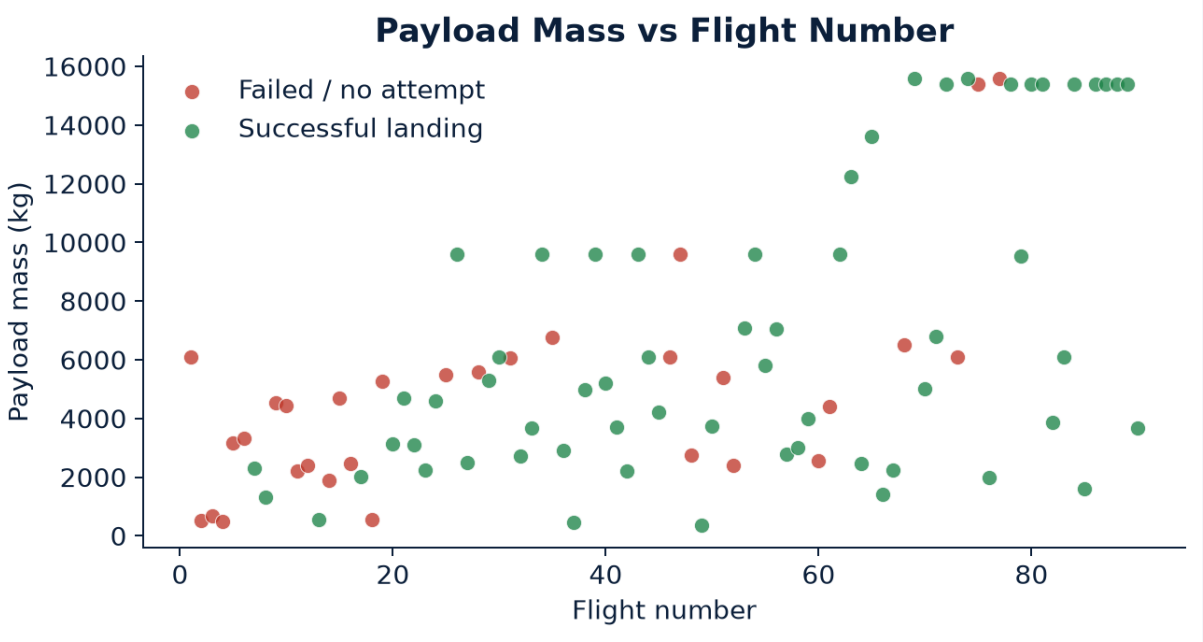
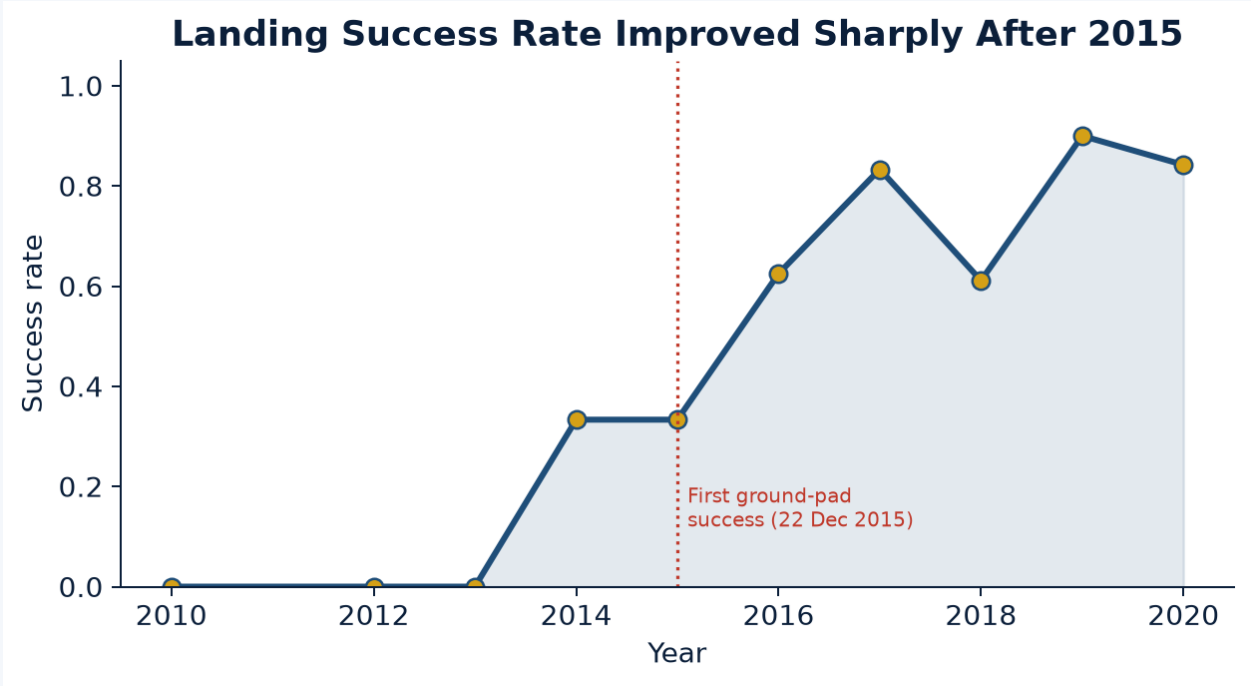
- CCAFS SLC 40: 55 launches (busiest). KSC LC 39A: 22. VAFB SLC 4E: 13 (polar / SSO corridor).
- KSC LC-39A shows the highest success rate; early CCAFS flights include the learning period before routine recovery.
- Later flight numbers at every site trend toward Class = 1 — experience and Block 5 reuse dominate site effects.

Orbit Type and Yearly Success

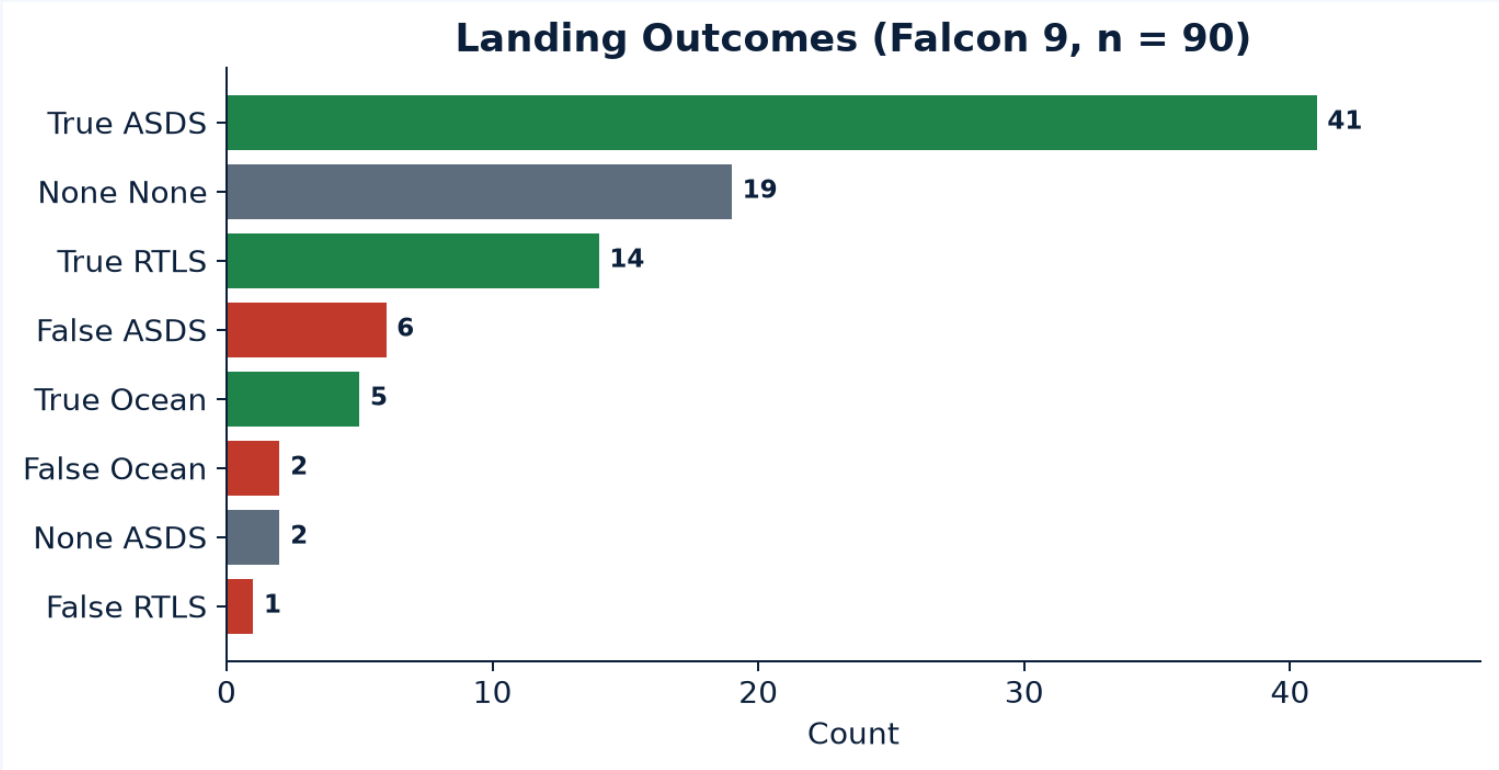


- GTO is a transfer orbit, not GEO. Only 1 launch is true GEO. GTO (27 flights) is large but not geostationary.
- SSO, VLEO, and several rare orbits (ES-L1, HEO, GEO) are near-perfect in this sample; GTO and ISS are mixed because of early expendable missions.
- Overall Class mean is 0.6667 (60 successes / 90). That average hides a steep time trend — next slide.

Time Trend and Payload Relationship



Landing Outcomes Drive the Label



True ASDS (drone ship) = 41 · True RTLS (ground pad) = 14 · True Ocean = 5 → 60 successful Class=1 landings.
None None (19) and other None/False outcomes form Class=0. False ASDS = 6 is the largest explicit failure mode.
Grid fins and legs are almost always present on successful RTLS/ASDS attempts — they are strong physical precursors of Class.

EDA with SQL — Key Query Answers

Question	SQL idea	Result
Unique launch sites	SELECT DISTINCT Launch_Site	CCAFS LC-40, SLC-40, KSC LC-39A, VAFB SLC-4E
5 CCA records	Launch_Site LIKE 'CCA%' LIMIT 5	First five Cape Canaveral rows
NASA (CRS) mass	SUM(mass) WHERE Customer = 'NASA (CRS)'	45,596 kg
F9 v1.1 average mass	AVG(mass) WHERE Booster LIKE 'F9 v1.1%'	2,534.67 kg
First ground-pad success	MIN(Date) for Success (ground pad)	2015-12-22
Drone ship, 4-6 t	Success (drone ship) AND 4000 < mass < 6000	B1022, B1026, B1021.2, B1031.2
Mission outcomes	COUNT(*) GROUP BY Mission_Outcome	100 successes, 1 in-flight failure
Max payload boosters	WHERE mass = (SELECT MAX(mass))	12 boosters at 15,600 kg

2015 Drone-Ship Failures and Outcome Ranking

2015 failure (drone ship)

`substr(Date,6,2) AS month, year = 2015`

01 · F9 v1.1 B1012 · CCAFS LC-40

04 · F9 v1.1 B1015 · CCAFS LC-40

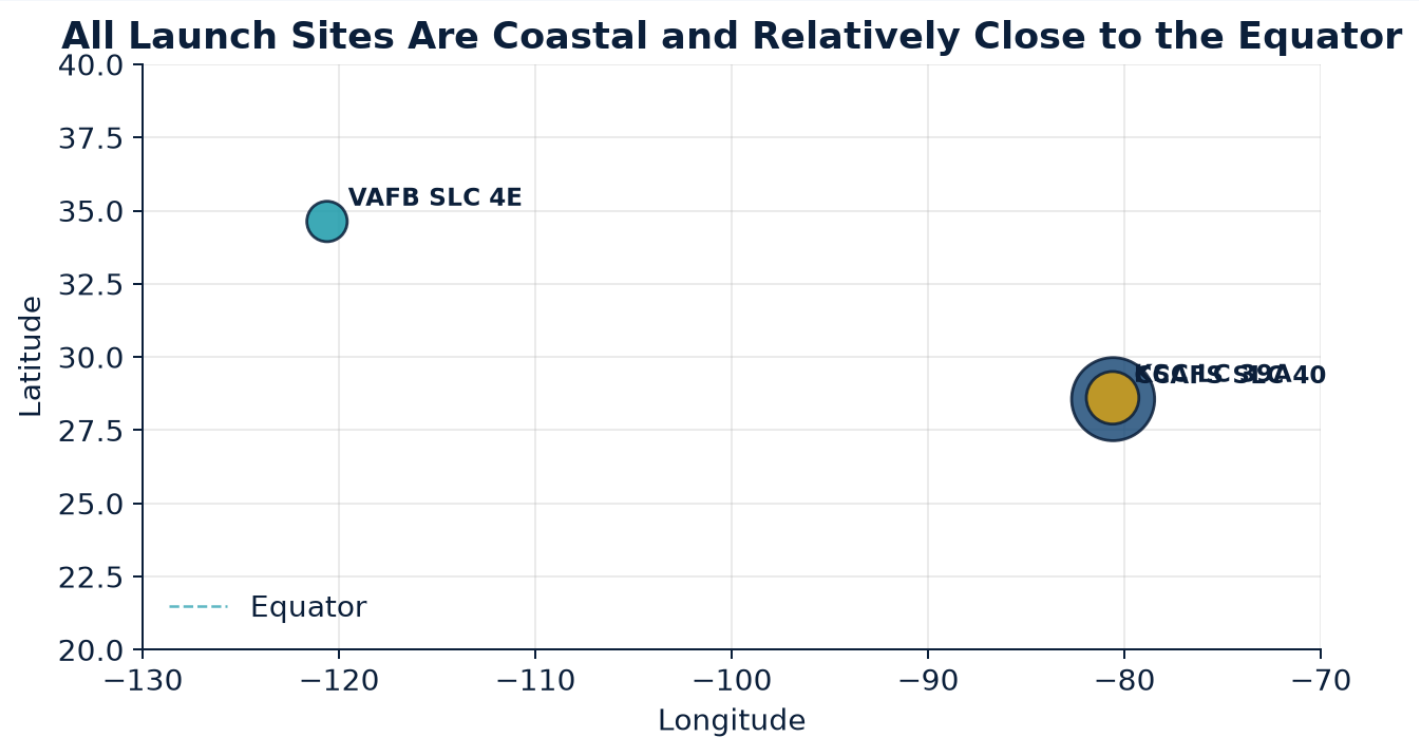
These are early ASDS attempts during the learning period before routine Block recovery.

Landing outcomes 2010-06-04 to 2017-03-20

- No attempt — 10
- Failure (drone ship) — 5
- Success (drone ship) — 5
- Controlled (ocean) — 3
- Success (ground pad) — 3
- Failure (parachute) — 2
- Uncontrolled (ocean) — 2
- Precluded (drone ship) — 1

- SQLite has no month-name function: use `substr(Date, 6, 2)` and `substr(Date, 0, 5)='2015'`.
- `LIMIT 20` (not `TOP/MAX`) is the correct way to cap rows. `MIN(payload)` is the correct aggregate for lightest mass.
- `COUNT(Mission_Outcome) GROUP BY Launch_Site` is the site outcome tally; `SUM` is the wrong aggregate for counts.
- Insight: through early 2017, 'No attempt' still dominated — recovery was optional. After 2017 it becomes the default mission profile.

Interactive Map Results (Folium)



Map construction

- `site_map.add_child(circle)` and `marker.add_to(map)` both work.
- Use `MarkerCluster` when many launches share a pad.
- Green marker = Class 1, red = Class 0.
- `MousePosition` to read lat/long; `PolyLine` for distance to coast/rail/highway/city.

All sites are on the coast, not inland cities: over-water trajectories, recovery ships, and abort modes.

Florida pads (CCAFS / KSC) sit close together (~28.6°N) for GTO/ISS. VAFB SLC-4E (~34.6°N, 120.6°W) serves polar/SSO.

Proximity pattern: near coast, rail, and highway; a safe buffer from dense city centers. Geography is a feature, not just a map.

Plotly Dash Dashboard Results

Application design

- dcc.Dropdown options hold the site list (All Sites + each pad). placeholder is hint text; value is the current selection.
- Callback outputs bind to unique component ids (pie chart, scatter). Dash does not guess the target from component name.
- Multiple Input components are allowed — site dropdown and payload RangeSlider jointly filter the figures.
- Pie: success vs failure for the selected site. Scatter: payload mass vs Class/outcome, colored by booster version.

All sites

Success pie matches the 60 / 30 Class split. Payload scatter shows later B5 boosters carrying 15.6 t Starlink stacks with Class=1.

KSC LC-39A

Highest site success share in the dashboard — consistent with the EDA bar chart. Good default for a reuse-cost bid.

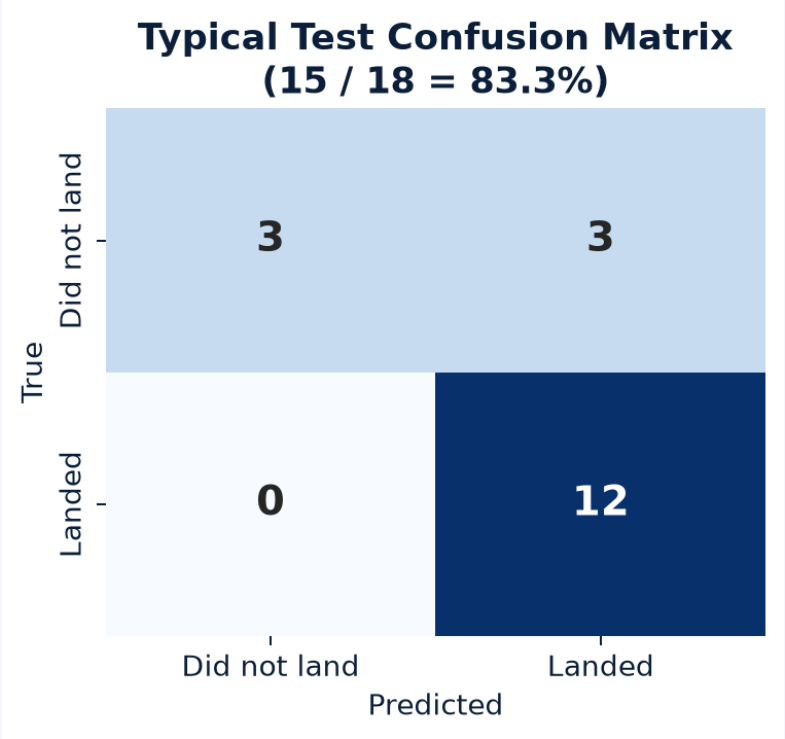
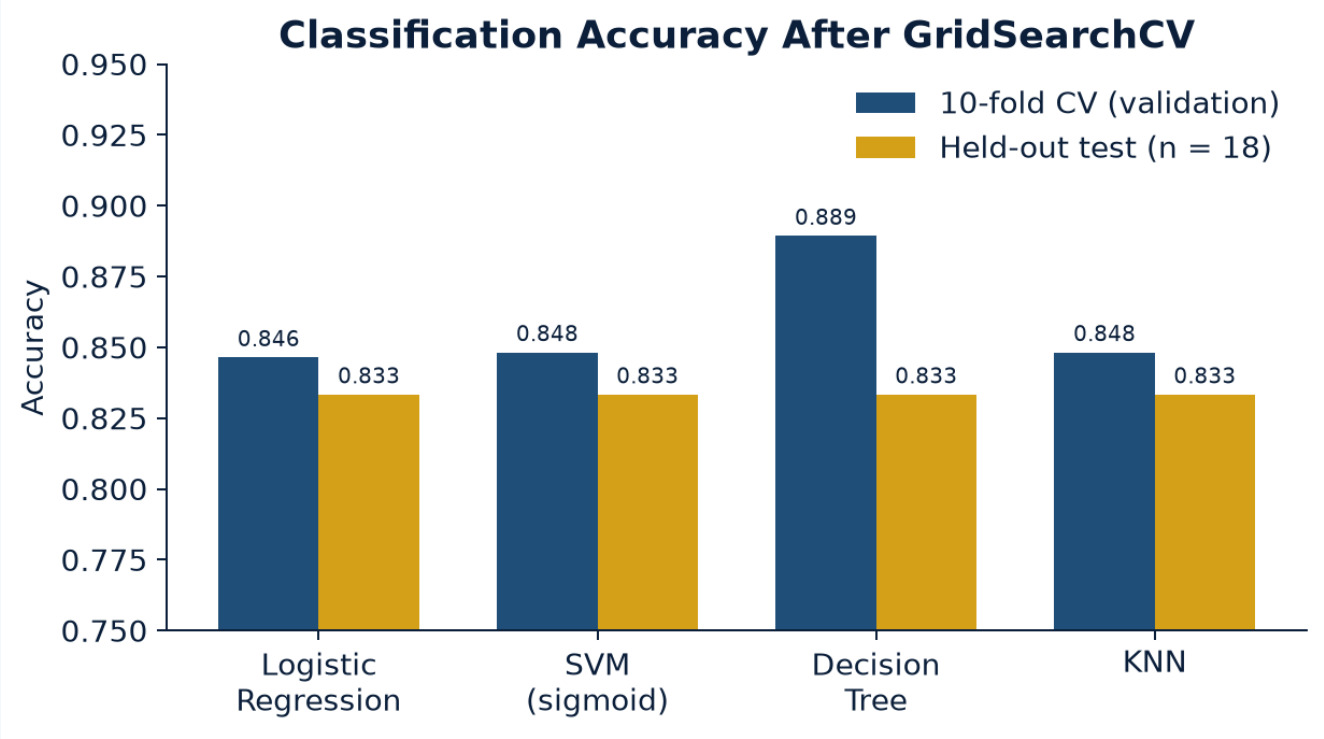
CCAFS SLC-40

Largest volume; mix of early failures and later Starlink successes. Filter by payload to see the post-2018 cluster.

VAFB SLC-4E

Fewer flights, polar/SSO. Success is high once ASDS recovery is in the mission plan.

Predictive Analysis Results



- Test set is only 18 records, so all four models can tie at 83.33% (15/18). Rank by GridSearchCV best_score_ on training folds.
- Best SVM kernel on validation: sigmoid (C=1.0, gamma≈0.0316, CV≈0.848). Logistic Regression best C=0.01.
- Decision Tree posted the highest CV accuracy (~0.889) and is the selected champion, with the same 0.833 test accuracy.
- Typical errors are false positives: predict land when the booster did not. That is the expensive mistake for a low-ball reusable bid.

Model Card — Hyperparameters and Scores

Model	Best hyperparameters	CV accuracy	Test accuracy
Logistic Regression	C=0.01, penalty=l2, solver=lbfgs	0.8464	0.8333
SVM	kernel=sigmoid, C=1.0, gamma≈0.0316	0.8482	0.8333
Decision Tree	highest CV; gini/entropy + depth/features grid	~0.8893*	0.8333
KNN	tuned n_neighbors / p / algorithm	0.8482	0.8333

* Decision tree CV varies slightly between runs (no fixed random_state in the lab grid). Use the first-run test score printed by tree_cv.score(X_test, Y_test).

Recommendation: ship the Decision Tree for interpretability (site, grid fins, legs, block, serial patterns) and keep Logistic Regression as a stable baseline. Do not over-claim 88% — the honest generalization estimate on this course test split is 83.3%.

Innovative Insights

LandingPad missingness is signal

26/90 pads are null. That is mostly 'no attempt', not broken data. Imputing a pad would leak a false recovery plan into the model.

The costly error is false 'land'

Predicting success and getting a splashdown prices the bid \$100M too cheap. Tune decision threshold toward precision on Class=1 for commercial bids.

Time is the hidden feature

A 2013 GTO mission and a 2020 Starlink VLEO mission are different processes. FlightNumber / Year explain more than Orbit alone.

VAFB is a different product

Polar/SSO from California is not interchangeable with Florida GTO. A startup should not copy SpaceX's Florida success rate for polar bids.

GTO $\neq$ GEO

27 GTO flights are transfer orbits. Only 1 GEO. Treating GTO as geostationary overstates high-energy success.

Reuse compounds

Serials such as B1049/B1051 fly many times. ReusedCount and Flights encode booster maturity; they should stay in the model.

Conclusion

- We assembled a reproducible Falcon 9 landing dataset (API + scrape + wrangling) with 90 flights and a 66.7% historical success rate.
- EDA shows the story is temporal and operational: after the 22 Dec 2015 ground-pad success, recovery became routine; KSC LC-39A and later Blocks lead.
- SQL independently confirms NASA CRS mass (45,596 kg), F9 v1.1 average mass (2,534.67 kg), max payload 15,600 kg on 12 boosters, and the 2015 ASDS failures.
- Folium and Dash make the geography and site filters usable: coasts, Florida vs VAFB, payload-success scatter.
- Four classifiers reach 83.3% test accuracy on 18 rows. Decision Tree wins validation (~88.9%). Use it, but price bids off 83% plus a false-positive penalty.
- Next: more recent flights, threshold tuning, and a cost model that converts predicted landing probability into a bid in dollars.

Sources, Assumptions, and Creativity Notes

- Primary data: IBM course mirrors of the SpaceX API, Wikipedia scrape, SpaceX.csv, dataset_part_2.csv / dataset_part_3.csv.
- Live api.spacexdata.com was unavailable (Cloudflare 525) during collection, so IBM static JSON + documented ID maps were used — same schema as the lab.
- Charts in this PDF were regenerated from dataset_part_2.csv so figures match the 90-row Falcon 9 analysis.
- Creativity: custom widescreen visual system, cost-gap chart, model card, geography-as-feature map, and an insights grid that goes past the Coursera template.
- GitHub: <https://github.com/lelandwrightofficial-alt/testrepo> — upload the notebooks listed on slide 2 if they are not already in that repo.
- This deck is the detailed technical report. An executive version is slides 1, 4, 18, and 21 only.